Sl. No	QUESTION	ANSWER
	MODULE 1	
1	A population refers to: A) A small part of data B) The entire group of individuals under study C) Only selected observations D) Random observations	
2	A sample is: A) Entire dataset B) Subset of a population C) Random variable D) Data entry method	
3	The main purpose of selecting **a** sample is to: A) Increase data size B) Reduce cost and time C) Avoid analysis D) Remove variables	
4	The process of selecting individuals from a population is called: A) Sampling B) Editing C) Coding D) Scaling	
5	Which of the following is **a** nominal level of measurement**?** A) Age B) Gender C) Temperature D) Height	
6	Which measurement level represents ranked data? A) Nominal B) Ordinal C) Interval D) Ratio	
7	Temperature measured in Celsius is an example of: A) Nominal scale B) Ordinal scale C) Interval scale D) Ratio scale	
8	Which scale has a true zero point**?** A) Nominal B) Ordinal C) Interval D) Ratio	
9	A rating scale is commonly used to measure: A) Attitudes or opinions B) Population size	

	C) Sample size D) Temperature	
10	The Likert scale is an example of: A) Nominal scale B) Rating scale C) Ratio scale D) Interval scale	
11	The variable that is manipulated or controlled in an experiment is called: A) Dependent variable B) Independent variable C) Constant variable D) Random variable	
12	The variable that changes as a result of the independent variable is called: A) Independent variable B) Dependent variable C) Nominal variable D) Ordinal variable	
13	In a study of study time and exam scores, study time is: A) Dependent variable B) Independent variable C) Nominal variable D) Ratio variable	
14	In IBM SPSS Statistics, datasets are typically opened using: A) File → Open → Data B) Edit → Data C) View → Data D) Analyze → Data	
15	Data in SPSS is displayed mainly in: A) Graph View B) Data View C) Model View D) Analysis View	
16	Viewing a few cases in SPSS helps to: A) Increase sample size B) Detect errors in data C) Delete data automatically D) Calculate statistics	
17	In SPSS, rows represent**:** A) Variables B) Cases or observations C) Charts D) Command	
18	The minimum value represents: A) Largest observation B) Smallest observation C) Average value	

	D) Median value	
19	Valid cases refer to: A) Missing observations B) Incorrect data C) Non-missing observations D) Deleted data	
20	Inconsistent responses in data usually indicate: A) Correct values B) Data entry errors C) Perfect responses D) Large samples	
21	Data checking is mainly performed to: A) Increase population size B) Identify errors and inconsistencies C) Remove variables D) Create graphs	
22	Missing values in IBM SPSS Statistics are represented as: A) Blank or defined missing codes B) Always zero C) Negative values D) Text values	
23	Missing values occur when: A) Data is complete B) Respondents skip questions C) Variables are deleted D) Charts are created	
24	The most common method to summarize categorical data is: A) Frequency table B) Regression C) Correlation D) T-test	
25	A frequency table shows: A) Mean values B) Number of occurrences of each category C) Variance D) Regression coefficients	
26	**A** bar chart is mainly used for: A) Continuous data B) Categorical data C) Correlation analysis D) Regression analysis	
27	In a bar chart, the height of the bar represents: A) Frequency or count B) Mean value C) Variance D) Median	

28	Standardizing the chart axis means: A) Changing chart color B) Adjusting scale for better comparison C) Deleting data D) Sorting data	
29	A pie chart represents: A) Relationship between variables B) Proportion of categories in a whole C) Mean values D) Variance	
30	In a pie chart, each slice represents: A) A variable B) A category's percentage or proportion C) Standard deviation D) Mean value	
	MODULE 2	
31	Exploratory Data Analysis (EDA) **is** mainly used to: A) Confirm hypotheses B) Explore and summarize data before formal analysis C) Remove variables D) Create models	
32	The main goal of EDA is to: A) Increase sample size B) Understand patterns and relationships in data C) Eliminate variables D) Avoid statistical tests	
33	Which scale of measurement has equal intervals but no true zero? A) Nominal B) Ordinal C) Interval D) Ratio	
34	If the tail of distribution extends to the LEFT**,** it is: A) Negatively skewed B) Positively skewed C) Symmetrical D) Uniform	
35	A histogram is used to display: A) Categorical data B) Continuous numerical data C) Text data D) Nominal variables	
36	In a histogram, the height of bars represents**:** A) Frequency B) Mean C) Variance D) Median	

37	Missing values occur when: A) Data is duplicated B) Data is not recorded for some observations C) Data is continuous D) Data is categorized	
38	One common method to handle missing data is: A) Ignoring all variables B) Mean substitution C) Increasing sample size randomly D) Removing all data	
39	A confidence interval provides: A) Exact population mean B) Range within which the population mean likely falls C) Maximum value D) Minimum value	
40	**A** 95% confidence interval means: A) Population mean is exactly known B) There is a 95% chance the interval contains the population mean C) Data is perfectly normal D) Sample size is fixed	
41	A normal distribution is: A) Skewed left B) Skewed right C) Symmetrical D) Random	
42	If the tail of distribution extends to the RIGHT**,** it is: A) Negatively skewed B) Positively skewed C) Symmetrical D) Uniform	
43	**A** stem-and-leaf plot is used to: A) Display numerical data while retaining original values B) Replace histograms C) Calculate regression D) Test hypotheses	
44	In a stem-and-leaf plot, the stem represents**:** A) First digit(s) of numbers B) Last digit only C) Mean value D) Frequency	
45	A box plot displays: A) Only mean values B) Five-number summary of data C) Regression lines D) Correlation	
46	The five-number summary includes:	

	A) Mean, Mode, Range, Variance, SD B) Minimum, Q1, Median, Q3, Maximum C) Mean, Median, Mode, Range, Variance D) Median, Mode, SD, Mean, Variance	
47	Saving an updated dataset ensures: A) Data is deleted B) Changes made to the dataset are preserved C) Data is duplicated automatically D) Variables are removed	
48	Increasing sample size generally: A) Decreases reliability B) Increases accuracy of estimates C) Removes variables D) Eliminates variance	
49	Testing for mean differences helps determine whether: A) Variables are categorical B) Two groups have significantly different means C) Sample size increases D) Data is normally distributed	
50	Before performing statistical tests, groups should be explored using: A) Graphs and descriptive statistics B) Regression models C) Factor analysis D) Structural equations	
51	When constructing a frequency table for interval scale data, what is the primary purpose of defining 'class intervals'? A. To group continuous data into manageable categories for better visualization of data distribution B. To eliminate the need for measures of central tendency C. To ensure that every data point is represented by unique bar D. To convert interval scale to nominal scale data	
52	In a Box & Whisker plot, what specific value does the line inside the box represent? A. Mean B. Mode C. Range D. Median	
53	Which measure of variability is most sensitive to extreme outliers in a dataset? A. Interquartile range B. Mode C. Mean Absolute Deviation D. Standard deviation	

54	What is the primary function of a Stem & Leaf Plot compared to a standard Histogram? A. It automatically calculates the confidence interval B. It provides more colorful visual representation C. It is used for categorical data D. It allows researcher to retain and see actual raw data values	
55	The main purpose of testing mean differences between groups is to determine whether: A. The sample size is large B. Two population means are significantly different C. Data is normally distributed D. The variance is zero	
56	In hypothesis testing, the assumption that there is no difference between groups is called: A. Alternative hypothesis B. Research hypothesis C. Null hypothesis D. Test hypothesis	
57	Error bars in graphs represent: A. Data entry errors B. Variability such as standard error or confidence interval C. Missing values D. Frequency counts	
58	WEAK deviations from the straight line in a normal probability plot indicate: A. Perfect normal distribution B. Non-normal distribution C. No variance D. Equal means	
59	Mean differences between groups can be visually displayed using: A. Bar charts B. Box plots C. Error bars D. All of the above	
60	Paired t-tests focus on: A. Differences within pairs B. Differences between independent groups C. Total sample mean D. Variance of groups	
MODULE 3		
61	The main purpose of a one-factor ANOVA is to test whether: A) All group means are equal B) Variances are zero C) The sample size is large D) Data is normally distributed	

62	In one-factor ANOVA, the null hypothesis (H_0) states that: A) At least one group mean is different B) All group means are equal C) Variances are unequal D) Sample size is sufficient	
63	In ANOVA terminology, a factor is: A) The dependent variable B) The grouping variable C) A numerical covariate D) The error term	
64	Before running ANOVA, researchers should check: A) Normality of the dependent variable B) Homogeneity of variances C) Outliers in groups D) All of the above	
65	In two-factor ANOVA, main effects refer to: A) The combined effect of both factors B) The effect of each factor independently C) Interaction effect only D) Variance within groups	
66	The F-test in two-factor ANOVA is used to test: A) Main effects of each factor B) Interaction effect C) Both A and B D) Only total mean differences	
67	If the interaction is not significant, researchers usually: A) Ignore main effects B) Interpret main effects independently C) Perform another ANOVA D) Only use post hoc tests	
68	Interaction plots are used to: A) Detect whether the effect of one factor depends on another B) Compare only main effects C) Calculate post hoc test values D) Measure sample size	
69	A study examines teaching method (3 levels) and study time (2 levels) on exam scores. How many factor combinations exist? A) 2 B) 3 C) 5 D) 6	
70	The interaction effect in two-factor ANOVA shows: A) How the combined levels of two factors influence the dependent variable B) The variance within groups C) The mean of each factor separately	

	D) The total sample size	
71	A two-factor ANOVA allows researchers to test: A) Only the effect of one factor B) The effect of two factors and their interaction C) Correlations between variables D) Differences in variances only	
72	ANOVA assumes: A) Independent observations B) Normally distributed residuals C) Homogeneity of variances D) All of the above	
73	If a study examines scores of buyer behaviour across three different consumer buying methods, the factor is: A) Exam score B) Buying method C) Sample size D) Post hoc test	
74	To visually compare group means in ANOVA, researchers often use: A) Line charts B) Bar charts with error bars C) Pie charts D) Scatter plots without groups	
75	Error bars in ANOVA graphs typically represent: A) Standard deviation or standard error B) Sample size C) Median D) Mode	